

23.2 Radioactive Decay

Question Paper

Course	CIE A Level Physics
Section	23. Nuclear Physics
Topic	23.2 Radioactive Decay
Difficulty	Easy

Time allowed: 60

Score: /44

Percentage: /100

Question 1a

Radioactive decay is often described as a spontaneous and random process.

State what is meant by

- (i)
radioactive decay [2]
- (ii)
a spontaneous process [2]
- (iii)
a random process [2]

[6 marks]

Question 1b

The graph in Fig. 1.1 shows the count rate of a radioactive substance measured by a Geiger-Müller tube.

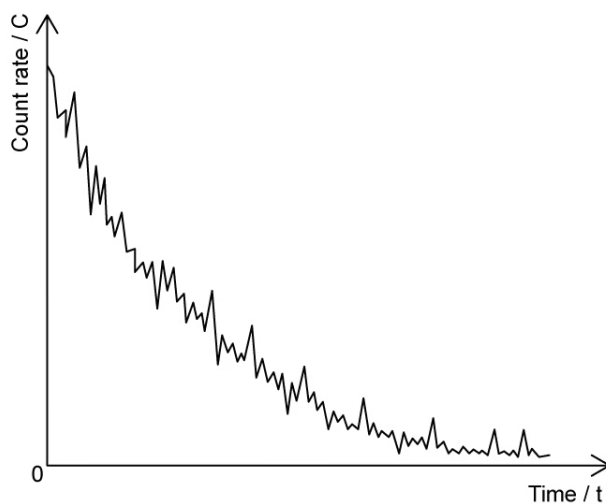


Fig. 1.1

State the feature of Fig. 1.1 which provides evidence for

- (i)
the random nature of radioactive decay
- (ii)
the spontaneous nature of radioactive decay

[1]

[1]

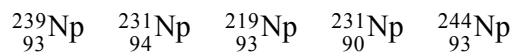
[2 marks]

Question 1c

The element neptunium has at least 24 isotopes. One of the isotopes is neptunium-231 (${}^{231}_{93}\text{Np}$), which has a half-life of 49 minutes.

(i)

State what is meant by *isotopes* and circle the possible isotopes of neptunium



[3]

(ii)

Define *half-life* and circle the correct expression for calculating the probability per second of decay of a nucleus of neptunium-231.

$$\frac{49}{0.693} \quad \frac{0.693}{49} \quad \frac{0.693}{49 \times 60} \quad \frac{49 \times 60}{0.693}$$

[3]

[6 marks]

Question 1d

Fig. 1.2 shows a diagram in which nucleon number A is plotted against proton number Z .

The diagram shows the position of three isotopes of neptunium and their respective decay products.

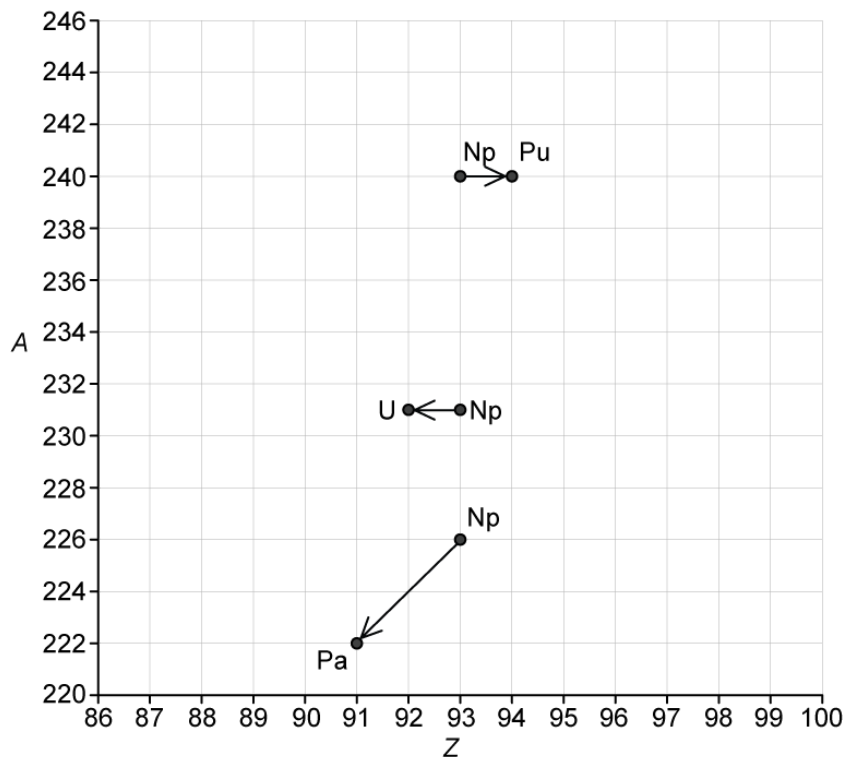
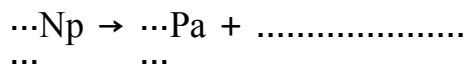
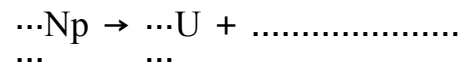
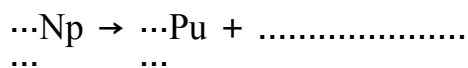


Fig. 1.2

Use Fig. 1.2 to complete the following nuclear decay equations



[3 marks]

Question 2a

Define half-life.

[2 marks]

Question 2b

A student investigates the half-life of technetium with time. This list shows the variables in the experiment.

time size of sample distance from the detector to the sample
same material for the sample activity of the sample

Using variables from the list, state

(i)
the independent variable

[1]

(ii)
the dependent variable

[1]

(iii)
the control variables for the experiment

[3]

[5 marks]

Question 2c

The experiment uses a variety of apparatus.

Draw lines on Table 1.1 below to match the apparatus with its correct use.

Table 1.1

Apparatus	Purpose
Long tongs	To detect the count rate of the radioactive source
Geiger-Müller tube	To measure the same interval of time for each reading
Secure holder for radioactive source	To handle the radioactive source at a distance
Stopwatch	To ensure the radioactive source remains in place. It is lined with lead to reduce unnecessary exposure

[4 marks]

Question 2d

Fig. 1.1 shows the graph that the student obtains from their results.

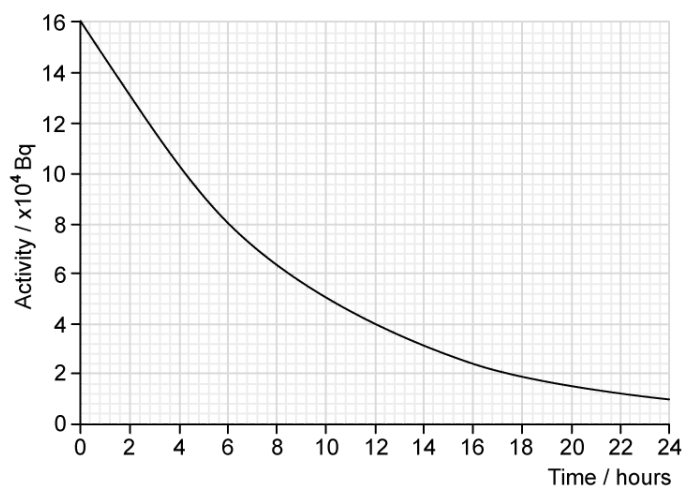


Fig. 1.1

Using Fig. 1.1, determine the half-life of the sample.

[3 marks]

Question 3a

An isotope of polonium-213 ($^{213}_{84}\text{Po}$) first decays into an isotope of lead-209 ($^{209}_{82}\text{Pb}$) and then decays into the stable isotope of bismuth (Bi).

Fig. 1.1 shows two arrows on a neutron number N against proton number Z chart to illustrate these two decays.

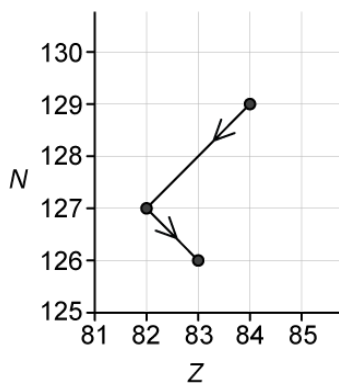
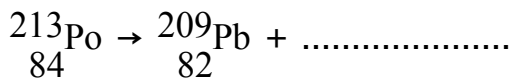


Fig. 1.1

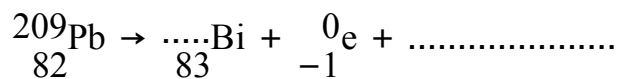
Use Fig. 1.1 to complete the nuclear decay equations for

- (i)
the polonium isotope



[1]

- (ii)
the lead isotope



[2]

[3 marks]

Question 3b

A pure sample of polonium-213 is produced in a research laboratory.

The half-life of ${}^{213}_{84}\text{Po}$ is very small compared with the half-life of ${}^{209}_{82}\text{Pb}$. After a very short time, the ionising radiation detected from the sample is mainly from the beta-minus decay of the lead-209 nuclei.

The half-life of ${}^{209}_{82}\text{Pb}$ is 3.3 hours.

(i)

State what is meant by the *decay constant* of a radioactive isotope.

[2]

(ii)

Calculate the decay constant, in s^{-1} , of lead-209.

[3]

[5 marks]

Question 3c

The activity of the sample of ${}^{209}_{82}\text{Pb}$ after 7.0 hours is 12 kBq.

Calculate the initial activity of the sample of polonium-213.

[2 marks]

Question 3d

(i)

State the relation between the activity A of a sample of a radioactive isotope containing N atoms and the decay constant λ of the isotope.

[1]

(ii)

Determine the number of lead-209 nuclei in the sample initially.

[2]

[3 marks]



Model Answers: Easy

1a

(a)

(i) Radioactive decay is defined as:

- The emission of a radioactive particle (alpha, beta or gamma); [1 mark]
- From a nucleus; [1 mark]

(ii) A spontaneous process is defined as:

- A decay which is unaffected by environmental changes; [1 mark]
- Such as temperature / pressure / chemical conditions; [1 mark]

(iii) A random process is defined as:

- The constant probability that a nucleus will decay (per unit time); [1 mark]
- (Because) it cannot be predicted which particular nucleus will decay next; [1 mark]

[Total: 6 marks]

1b

(b)

(i) The graph demonstrates the random nature of radioactive decay because:

- There are fluctuations in the curve **OR** the curve is not smooth; [1 mark]

(ii) The graph demonstrates the spontaneous nature of radioactive decay because:

- The time of decay (of a nucleus) cannot be predicted **OR** the nucleus has a constant probability of decaying in a given time; [1 mark]

[Total: 2 marks]

It is not possible to determine exactly when radioactive nuclei will decay. It could be in the next minute, or it might not be for another 10 000 years or more!

All we can do is look at the general **pattern** across a large number of radioactive nuclei which provides us with the half-life. The decay of an individual nucleus is **not** dependent on the others in the sample.

The fluctuations in count rate, as shown on the graph, indicate that sometimes slightly more nuclei will decay in a period of time and sometimes slightly fewer!

1c

(c)

(i) Isotopes of an element are:

- Nuclei which have the same proton number / atomic number / number of protons; [1 mark]
- But a different neutron number / number of neutrons; [1 mark]

Possible isotopes of neptunium:

- ${}_{93}^{239}\text{Np}$, ${}_{93}^{219}\text{Np}$, ${}_{93}^{244}\text{Np}$ (all three circled); [1 mark]

(ii) Half-life is defined as:

- The time it takes for the initial activity (of a source) / number of nuclei (in a source); [1 mark]
- To reduce to one-half of its initial value; [1 mark]

The correct expression is:

- $(\lambda =) \frac{0.693}{49 \times 60}$ [1 mark]

[Total: 6 marks]

For part (i), make sure to look carefully at each option - the two incorrect options ${}_{94}^{231}\text{Np}$ and ${}_{90}^{231}\text{Np}$ have atomic (proton) numbers of 94 and 90 which means they would be completely **different elements**

For part (ii), you needed to realise that the probability per second of decay of a nucleus is the definition of **decay constant** and that you need to multiply the **half-life by 60** to convert it into seconds

1d

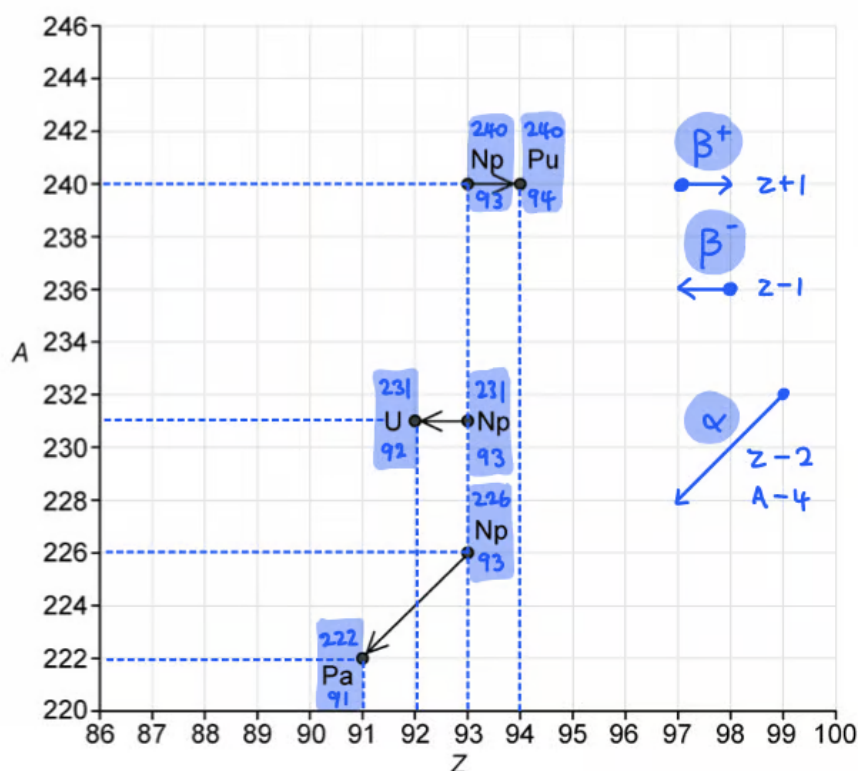
(d) The correct nuclear decay equations are:

- ${}_{93}^{240}\text{Np} \rightarrow {}_{94}^{240}\text{Pu} + {}_{-1}^0\beta + \bar{\nu}$; [1 mark]
- ${}_{93}^{231}\text{Np} \rightarrow {}_{92}^{231}\text{U} + {}_{+1}^0\beta + \nu$; [1 mark]
- ${}_{93}^{226}\text{Np} \rightarrow {}_{91}^{222}\text{Pa} + {}_2^4\alpha$; [1 mark]

[Total: 3 marks]

Taking some time to annotate the graph can help to avoid silly errors - make sure you understand that **beta-minus** decay will result in a nuclide with one less proton, and **beta-plus** decay will result in a nuclide with one extra proton

Alpha decay is easily recognisable as it results in a decrease in **both** proton and nucleon number



2a

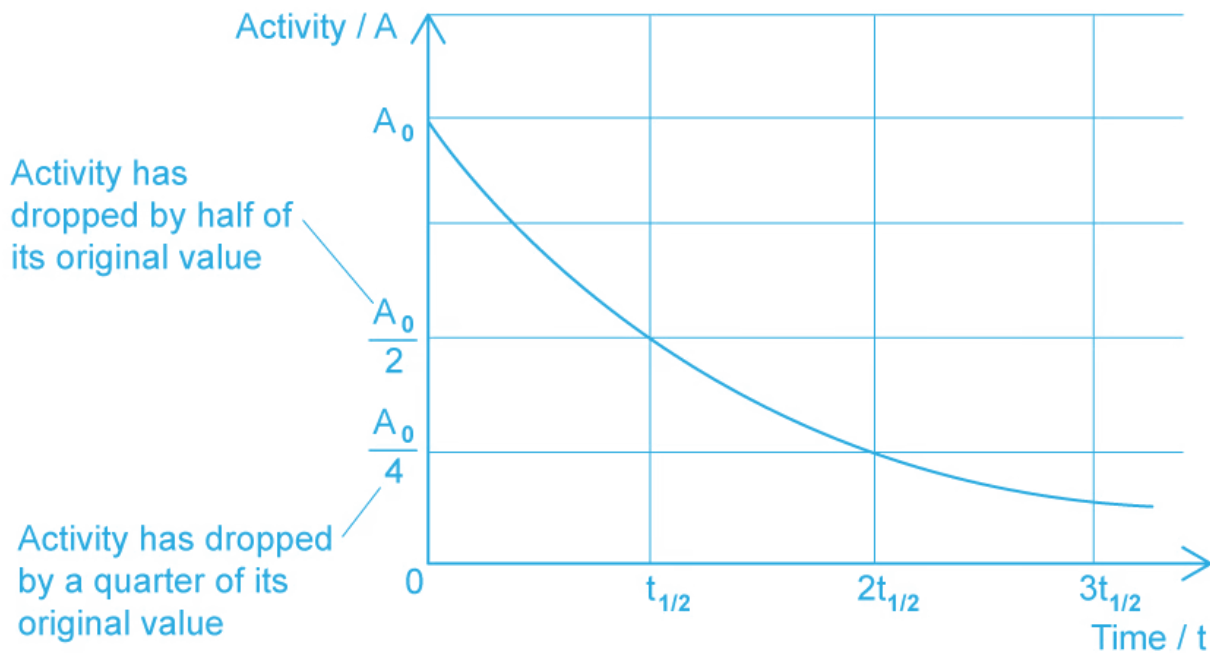
(a) The definition of half-life is:

- The time taken for the number of nuclei / the activity (of the source); [1 mark]
- To decay / reduce by half; [1 mark]

[Total: 2 marks]

The time it takes for the activity of the sample to decrease from 100% to 50% is the half-life. It is the same length of time as it would take to decrease from 50% activity to 25% activity. The half-life is constant for a particular isotope.

The diagram shows how the activity of a radioactive sample changes over time. Each time the original activity halves, another half-life has passed.



The diagram shows how the activity of a radioactive sample changes over time. Each time the original activity halves, another half-life has passed.

2b

(b)

(i) The independent variable is:

- Time; [1 mark]

(ii) The dependent variable is:

- Activity of the sample; [1 mark]

(iii) The control variables are:

- Size of sample; [1 mark]
- Distance from the detector to the sample; [1 mark]
- Same material for the sample; [1 mark]

[Total: 5 marks]

2c

(c) The correct purpose of each piece of apparatus is as follows:

Apparatus
Long tongs

Purpose
To detect the count rate of

- Time; [1 mark]

(ii) The dependent variable is:

- Activity of the sample; [1 mark]

(iii) The control variables are:

- Size of sample; [1 mark]
- Distance from the detector to the sample; [1 mark]
- Same material for the sample; [1 mark]

[Total: 5 marks]

2c

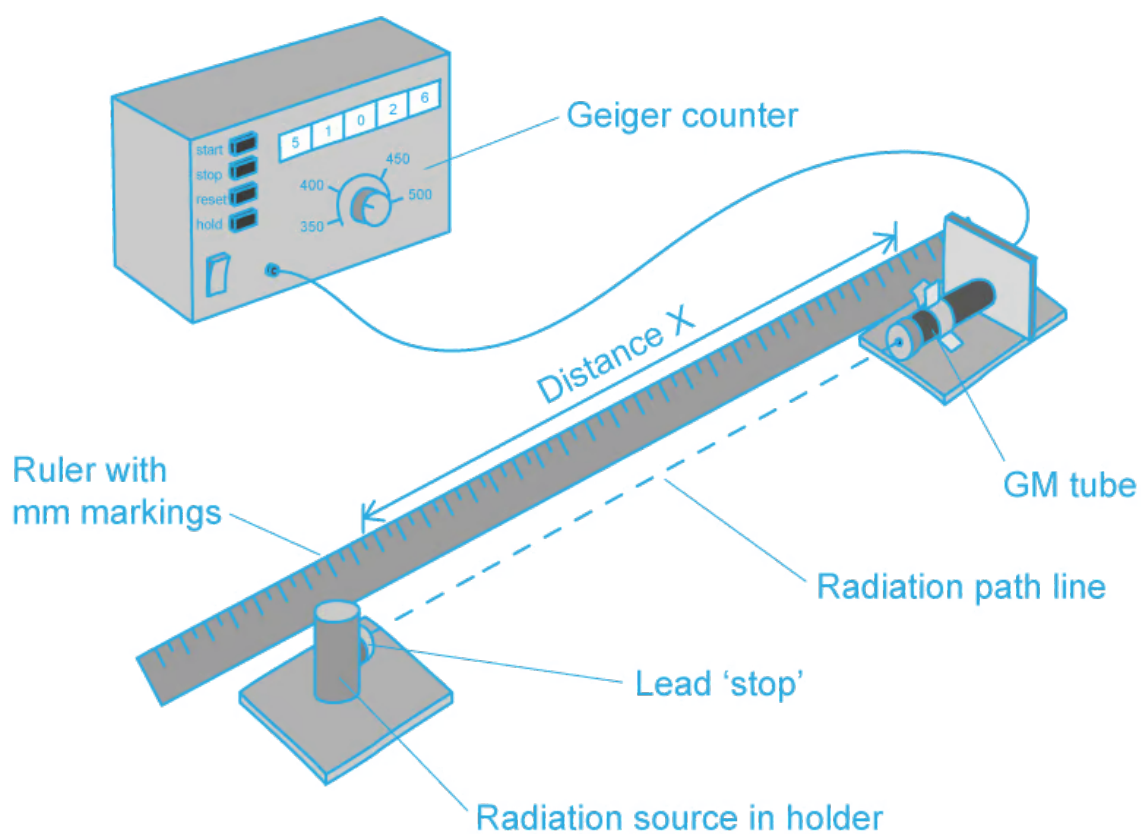
(c) The correct purpose of each piece of apparatus is as follows:

Apparatus	Purpose
Long tongs	To detect the count rate of the radioactive source
Geiger-Müller tube	To measure the same interval of time for each reading
Secure holder for radioactive source	To handle the radioactive source at a distance
Stopwatch	To ensure the radioactive source remains in place. It is lined with lead to reduce unnecessary exposure

- 1 mark for each correct line drawn

[Total: 4 marks]

The set up for the experiment is shown below:



2d

(d) Determine the half-life of the sample:



- (From the graph) initial activity: $A_0 = 16 \times 10^4 \text{ Bq}$ [1 mark]
- Half the initial activity = $\frac{1}{2} A_0 = 8 \times 10^4 \text{ Bq}$ [1 mark]
- (From the graph) half-life: $t_{1/2} = 6 \text{ hours}$ [1 mark]

[Total: 3 marks]

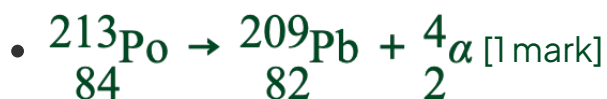
The definition of half-life can help you remember how to approach questions involving graphs:

The time taken for half the undecayed nuclei to decay or the activity of a source to decay by half

3a

(a)

(i) The nuclear decay equation for the polonium isotope is:



(ii) The nuclear decay equation for the lead isotope is:



[Total: 3 marks]

3b

(b)

(i) Decay constant is defined as:

EITHER

- The probability of decay (of a nucleus) / the fraction of the number of nuclei in a sample that decay; **[1 mark]**
- Per unit time; **[1 mark]**

OR

- $\lambda = -\left(\frac{dN}{dt}\right)\frac{1}{N}$; **[1 mark]**
- Where $-\left(\frac{dN}{dt}\right)$ is the rate of decay of nuclei and N is the number of nuclei in the sample; **[1 mark]**

(ii) Calculate the decay constant of lead-209:

List the known quantities:

- Half-life of lead-209, $t_{1/2} = 3.3 \text{ hours} = 3.3 \times 60 \times 60 = 11\,880 \text{ s}$ **[1 mark]**

Calculate the decay constant, λ :

- Decay constant: $\lambda = \frac{0.693}{t_{1/2}}$

- $\lambda = \frac{0.693}{11\,880}$ [1 mark]
- $\lambda = 5.83 \times 10^{-5} = 5.8 \times 10^{-5} \text{ s}^{-1}$ (2 s.f.) [1 mark]

[Total: 5 marks]

3c

(c) Calculate the initial activity of the sample of polonium-213:

List the known quantities:

- Activity after 7 hours, $A = 12 \text{ kBq} = 12 \times 10^3 \text{ Bq}$
- Time, $t = 7 \text{ hours} = 7 \times 60 \times 60 = 25\,200 \text{ s}$
- Decay constant of lead-209, $\lambda = 5.83 \times 10^{-5} \text{ s}^{-1}$ (from **(b)**)

Calculate the initial activity using the exponential decay equation:

- Exponential decay: $A = A_0 e^{-\lambda t} \Rightarrow A_0 = \frac{A}{e^{-\lambda t}}$
- $A_0 = \frac{12 \times 10^3}{e^{-(5.83 \times 10^{-5}) \times (25\,200)}}$ [1 mark]
- $A_0 = 5.21 \times 10^4 = 5.2 \times 10^4 \text{ Bq}$ (2 s.f.) [1 mark]

[Total: 2 marks]

3d

(d)

(i) State the relationship between A , N and λ :

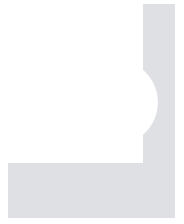
- $A = \lambda N$ [1 mark]

(ii) Determine the number of lead-209 nuclei in this sample

List the known quantities:

- Initial activity, $A_0 = 5.21 \times 10^4 \text{ Bq}$ (from **(c)**)
- Decay constant, $\lambda = 5.83 \times 10^{-5} \text{ s}^{-1}$ (from **(b)**)

Calculate the initial number of lead nuclei:



(d)

(i) State the relationship between A , N and λ :

- $A = \lambda N$ [1 mark]

(ii) Determine the number of lead-209 nuclei in this sample

List the known quantities:

- Initial activity, $A_0 = 5.21 \times 10^4 \text{ Bq}$ (from (c))
- Decay constant, $\lambda = 5.83 \times 10^{-5} \text{ s}^{-1}$ (from (b))

Calculate the initial number of lead nuclei:

- Rearrange: $A_0 = \lambda N_0 \Rightarrow N_0 = \frac{A_0}{\lambda}$
- $N_0 = \frac{5.21 \times 10^4}{5.83 \times 10^{-5}}$ [1 mark]
- $N_0 = 8.94 \times 10^8 = 8.9 \times 10^8$ (2 s.f.) [1 mark]

[Total: 3 marks]

Units are extremely important throughout this question!

When using the equation $x = x_0 e^{-\lambda t}$, it doesn't matter which unit of time you use for λ and t as long as they have the **same units** (e.g. s^{-1} and s or hr^{-1} and hr)

However, for the equation $A = \lambda N$, A is in Bq, or s^{-1} , therefore λ must be in s^{-1}